SUBHRANSU PRIYARANJAN NAYAK

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Education

Indian Institute of Technology (IIT) Bhubaneswar — **2021 – 2026**
Integrated B.Tech in Mechanical System Design Engineering — ***CGPA: 8.3*** | *Odisha, India*

Achievements

- **Competitive Programming & DSA:** Solved over **1,500+ problems** | LeetCode: Peak Rating **1934** (Top 4%, **1000+** Problems solved) | CodeChef: Peak Rating **1682** (**3-Star**) | Codeforces: Peak Rating **1345** (**Pupil**) (**200+** Problems solved).
- **Bronze Medalist: Inter IIT Tech Meet 10.0 (2022):** Secured **3rd place** for IIT Bhubaneswar across **23 IITs** [Certificate].
- **Intra-IIT Bhubaneswar Hackathon: 3rd place** in Game Dev (UE5) [Code] | **3rd place** in WebDev (ERP System) [Code].
- **IIT Bhubaneswar's Official Mobile App:** Contributed to the development of a **cross-platform mobile app** [Code] for GC 2024.

Experience

Data Engineering Intern | *HealthyDay [Offer Letter]* *Hybrid, India* | **Nov 2025 – Present** ***(3Mo. + 6Mo. Extended)***
- Engineered a **BigQuery**-based data processing layer for a platform serving **600K+ registered students**, integrating Firestore and relational operational data by developing **50+ reusable SQL views** with CTEs, window functions, JSON parsing, regex classification, and temporal deduplication.
- Designed and optimized **150+ complex SQL queries** and data pipelines across user **attendance, referral, subscription, payment, and messaging workflows** using incremental aggregation strategies and query tuning.
- Built **end-to-end data pipelines** unifying heterogeneous sources **(Firestore, Google/Meta Ads, WhatsApp/AiSensy messaging logs, transactional databases)**, implementing entity matching, and event sequencing across system boundaries.
- Developed automated data models and analytical views powering **Looker Studio** dashboards, implementing **funnel/event processing, cohort retention analysis, referral tracking, customer support metrics, and AI-classifier log monitoring**.

Software/AI Engineering Intern | *Hanyaa Auto Technologies [Experience Letter]* *Hyderabad, India* | **Jun 2025 – Jul 2025**
- Engineered an **AI text-to-animated-video narration pipeline** for Indian languages (EN, HI, TE) by fine-tuning **Dia-TTS (1.6B)**, optimizing training with dynamic LR scheduling and gradient clipping to eliminate robotic artifacts.
- Architected a long-form audio generation engine with sentence chunking (**NLTK**), input text normalization, failure-handling retry logic, and seed-locked speaker consistency (**IS11 tag**) for coherent synthesis.
- Built an interactive **Streamlit UI** integrating **Gemini 2.5 Flash TTS API** alongside **proprietary media pipelines(Hanyaa's)**, enabling automated multilingual narration generation.

Projects

DynamoCore [Code] | **A Highly Available Key-Value Storage**| **Java, RocksDB, Maven, Gossip Protocol, Vector Clock**
- Implemented the Amazon Dynamo paper (2007) end-to-end using Core Java and its multithreading features, built a distributed **AP key-value store** with asynchronous quorum consensus (N=3, W=2, R=2) via non-blocking `CompletableFuture` pipelines, achieving **12 ms p99 write latency** under concurrent load.
- Implemented a **consistent hashing ring**(150 virtual nodes/host, MurmurHash3) with $O(\log N)$ key routing, achieved **sub-4% load distribution deviation** and bounded node-join/leave data migration to K/N, eliminating full-ring reshuffling on topology changes.
- Sustained **100% Writes** during primary replica failures using **sloppy quorums**, hinted handoffs, and a **P2P Gossip failure detector** (ALIVE/SUSPECT/DEAD) with automated background recovery replay, zero writes lost during simulated node outages.
- Eliminated the full-dataset reconciliation scans by implementing **Merkle Tree anti-entropy** with $O(1)$ root-hash validation (**6 ms p99**) – used immutable **Vector Clocks** for causal conflict detection, enabling targeted replica repair without global consistency locks.

UrbanEats [Code] | **Event-Driven Food Delivery Platform** | **React, Node, TypeScript, RabbitMQ, Socket.io, Docker**
- Decoupled core domains into **6 autonomous Node.js/TypeScript microservices** communicating via HTTP and asynchronous **RabbitMQ AMQP message queues**, ensuring fault tolerance and zero-downtime service autonomy.
- Built a real-time courier dispatch using **MongoDB `2dsphere` spatial indexing** and **`$nearSphere`** queries to match the nearest available courier within **500 m** – streamed live GPS telemetry at 10-second intervals to React/Leaflet clients via **Socket.IO** room-based routing.
- Automated abandoned cart with **MongoDB TTL indexes**(15-mins) and secured payment finalization with **Razorpay HMAC SHA-256** signature validation – used **RabbitMQ** to atomically persist confirmed transactions, preventing double-processing on retries.
- Optimized the media pipelines by streaming Multer in-memory buffers as **Base64 DataURIs to Cloudinary CDN** for zero-disk asset uploads, utilizing native MongoDB drivers to eliminate ODM overhead during bulk KYC verification workflows.

Enterprise RAG Orchestrator [Code] | **Agentic RAG & Text-to-SQL Engine** | **FastAPI, LangGraph, Qdrant, Redis**
- Built a **stateful multi-path reasoning engine** with **LangGraph** and PostgreSQL checkpointers – dynamically routes queries across RAG, Text-to-SQL, and Hybrid paths; implemented **Corrective RAG (CRAG)** web fallbacks and **Self-RAG** reflection loops to reduce hallucinations across a **95% noise document corpus**.
- Engineered a multi-stage retrieval pipeline with **Qdrant hybrid search (Dense + Sparse TF-IDF via RRF), HyDE expansion, and Cross-Encoder reranking (Top-20 to Top-5)**, backed by a 5-tier cryptographic **Redis cache**.
- Developed an **AST-guarded Text-to-SQL engine** over 7 PostgreSQL 16 tables deployed on K8s – enforced **SELECT-only** execution policies and integrated LangGraph **`interrupt()` HITL** approval gates to block AI-generated mutation queries before execution.
- Hardened a **9-layer defense architecture** across FastAPI endpoints with ML prompt injection scanning (`llm-guard`), **XML spotlighting, bidirectional PII redaction, token budgeting, and Ragas benchmarking**.

Additional Projects: transaction-ledger-backend *(Idempotency Keys, Atomic Transactions, Ledger Model, JWT Blacklisting)* | realtime-collaborative-editor *(Liveblocks, Convex, Zustand, ProseMirror, Tiptap, Clerk, Zod, TypeScript, Next.js, React Hook Form)*

Skills

- **Languages:** Java, Python, C, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Core CS & System Design:** Data Structures & Algorithms, Operating Systems, Multithreading, Networking, DBMS, Distributed Systems, Microservices Architecture, High-Level Design, Design Patterns, OOPs, API Design, Data Modelling
- **Web & Frameworks:** React.js, Next.js, Node.js, Express.js, React Native, FastAPI, Streamlit, Socket.io, Tailwind, REST APIs
- **Databases & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Firebase, Docker, RabbitMQ, Git, Linux, CI/CD
- **AI/ML & GenAI:** PyTorch, TensorFlow, LangGraph, RAG, Hugging Face, Transformers, Scikit-learn, XGBoost, NLTK

Certifications

System Design (Grokking) (2026), System Design Basics (2026), Advanced SQL Querying (2025), SQL (2023), Algorithms & CP (2025), Java In-Depth (2023), Deep Learning (2023), Machine Learning (2023), The React.js Guide (2025), The React Native Guide (2025)